

Follicular Circadian Reproductive Rhythms: A Mathematical Model

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Abstract

The female reproductive endocrine system primarily consists of gonadotropin releasing hormone ($GnRH$), follicle-stimulating hormone (FSH), luteinizing hormone (LH), estrogen (estradiol or E_2), and progesterone (P_4). Pulses of $GnRH$ prompt the production of LH and FSH , which in turn stimulates estradiol and progesterone growth. While a monthly model of these hormones is widely known, our bodies have a circadian rhythm, and certain hormones fluctuate on a daily basis. Studies suggest that these hormones more closely follow a circadian rhythm during the follicular phase. Thus, this research paper will map the fluctuation of the hormones most involved in the menstrual cycle on a minute-by-minute scale, under the influence of melatonin, to demonstrate circadian rhythm during the follicular phase.

Keywords: Gonadotropin-releasing hormone, Luteinizing Hormone, Follicle Stimulating Hormone, Circadian Rhythm, Melatonin

1 INTRODUCTION

It is well known that female reproductive hormones fluctuate on a monthly basis. There are preexisting mathematical models that demonstrate the behavior of these hormones over a monthly time period. These models are especially useful for studying hormone therapies, fertility, and other conditions pertaining to the disruption of these reproductive hormones [8]. Looking at a smaller time scale, namely a single day, we see that each cell in our body has a circadian clock [10]. This biological clock has led us to consider how these hormones (particularly follicle-stimulating hormone (FSH), luteinizing hormone (LH), estrogen (E_2), and progesterone (P_4)), fluctuate on a daily basis. Through reviewing the literature, we have found numerous studies done on the circadian rhythm (and/or sleep) and the menstrual cycle [[1], [11], [13], [14], [12], [16]], as well as a model for these hormones on the scale of a minute during the early follicular and mid-luteal phases of the menstrual cycle [15]. Our goal is to use the graphs and data from Rahman et al. as a framework to modify the differential equations from Rasgon et al. to create a mathematical model for the daily fluctuation of these reproductive hormones during the follicular phase of the menstrual cycle. This model could then be used to analyze mood disorders [15], further understand the interaction of sleep and the phases of the menstrual cycle, and study exogenous hormone treatments on a daily level. In this paper, we will outline the biological processes involved in the secretion of gonadotropin releasing hormone ($GnRH$), FSH , LH , E_2 , and P_4 as well as the interaction of melatonin in this system. We will then discuss our methods for setting up our model and delve into the applications of this model.

2 Background

2.1 Circadian Rhythm

The circadian rhythm is the internal timekeeping system in our bodies. Circadian rhythms occur independent of external cues like light and other environmental factors [17]. To study the circadian rhythm of LH , FSH , E_2 , and P_4 we are looking at the triggering affects of melatonin. Melatonin regulates many of the biological clocks in our bodies. The central timekeeper, located in the brain, is called the suprachiasmatic nucleus (SCN). The SCN is located in the anterior part of the hypothalamus and is made up of roughly 20000 neurons.

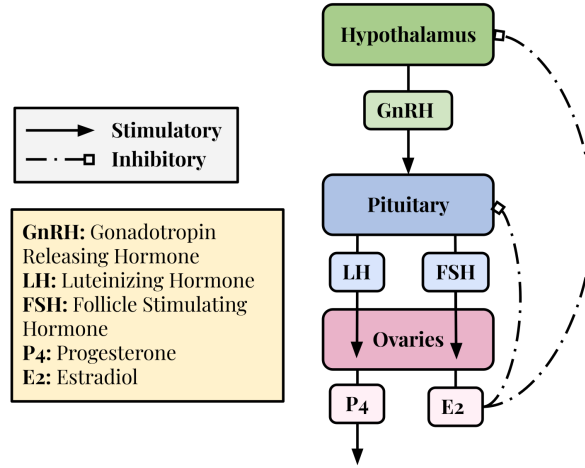
2.2 Menstrual cycle

There are four different phases of the menstrual cycle. The cycle begins with the follicular phase, which builds up to ovulation. This is followed by the luteal phase, which leads to the final phase, menstruation. Each of these phases is characterized by the rise and fall of the 4 primary hormones: pituitary hormones produced in the pituitary gland, the lutenizing hormone (LH) and the follicle-stimulating hormone (FSH), and the sex steroids produced in the ovaries, estradiol (E_2) and progesterone (P_4). We will primarily focus on the follicular phase, as this is the phase during which

these hormones show the most daily rhythm. During the follicular phase, FSH rises and LH remains low. E_2 and P_4 levels start off low, but E_2 rises by the midfollicular phase. P_4 remains relatively low with a slight increase towards the end of the follicular phase [7].

2.3 Hypothalamic-Pituitary-Ovarian Axis

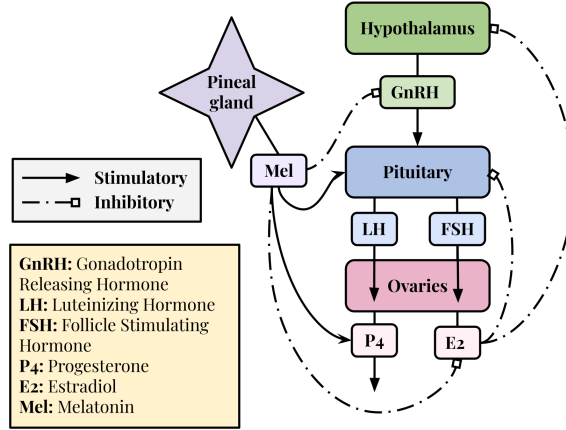
The hypothalamic-pituitary-ovarian (HPO) axis consists of the hypothalamus, the pituitary gland, and the ovaries. In the hypothalamus, located in the brain, a neuropeptide kisspeptin stimulates the production of the gonadotropin-releasing hormone ($GnRH$) [4]. $GnRH$ primarily signals the anterior pituitary gland to release LH and FSH .



Although both of these pituitary hormones play a role in the secretion of both E_2 and P_4 , we are focused on the triggering effect of FSH on E_2 and of LH on P_4 [9][6]. Dependent on level, E_2 inhibits $GnRH$, LH , and FSH , giving negative feedback. In addition P_4 has a similar effect, inhibiting $GnRH$, LH , and FSH . However, due to P_4 's lower levels during the follicular phase, these effects are negligible and it will be functionally seen as though P_4 exits the system.

2.4 Melatonin

Melatonin is produced in the pineal gland, from which it is released into the circulatory system. Melatonin inhibits the production of kisspeptin, the hypothalamic neuropeptide responsible for stimulating the release of $GnRH$. As the concentration of melatonin increases, $GnRH$ pulse frequency decreases. A heightened level of melatonin also increases the sensitivity of both LH and FSH [3]. In addition to melatonin's direct effect on $GnRH$ it also has a direct effect on P_4 . Melatonin stimulates the production of P_4 through the "upregulation of StAR expression" [5]. Lastly, Melatonin inhibits LH and FSH stimulated E_2 production [2].



3 Methods

Our set of equations was run by MATLAB, using function ode4 with a fixed timestep of 0.25 minutes. We used a fixed time-step method over one of MATLAB's built in variable step size solvers, since *GnRH* pulses last for only 0.5 minutes, and can be easily missed by a larger step size.

Our concentrations are measured in *IU/L* for LH and FSH, *nmol/L* for *P4*, and *pmol/L* for *E2*.

We primarily used Rasgon et al. as a basis for our model without the influence of melatonin. Using Rasgon et al. allowed us to capture the interactions between the hormones *GnRH*, *LH*, *FSH*, *E2*, and *P4* on a minute by minute scale, demonstrating how *LH* and *FSH* react with each spike of *GnRH*. Within our final set of differential equations, the following methods were taken from Rasgon et al.

GnRH is represented by default as an hourly pulse; period and amplitude of these pulses are inversely related to the level of estrogen. Our pulse function here is written to replicate the built-in pulse function of differential equation solver Berkeley Madonna, used by Rasgon et al. to run their system.

For *LH*, the equation contains a positive term that uses a multiplier on the *GnRH* value. This multiplier is dependent on estradiol so that in an excess of estradiol, *LH* is less sensitive to the pulses of *GnRH*.

Since we know that *FSH* is less sensitive than *LH* in reactivity to *GnRH* pulses, the effect of these pulses is given by an upward sigmoidal curve to represent *GnRH*'s positive feedback on *LH*. In addition, *FSH* is inhibited by *E2*, and this is represented by a downward sigmoidal feedback curve. Finally, *FSH* inhibits itself by way of a metabolic clearance term, similar to that of *LH*.

Once *GnRH* pulses affect *LH* and *FSH*, the increase in these hormones then stimulates E_2 and P_4 .

The differential equation for E_2 relies on positive feedback from *FSH*, given by an upward sigmoidal curve, and a metabolic clearance term. To account for the melatonin multiplier in the E_2 differential equation we must find a way to model the relationship between melatonin and E_2 . To accomplish this we plotted E_2 against melatonin with melatonin as the independent variable. After plotting we estimated a best-fit line (specifically a Gaussian function) for the data and used this as our M_E multiplier.

Lastly, the differential equation for P_4 is dependent on positive influence from *LH* and negative influence from P_4 's metabolic clearance.

From here, we were able to incorporate melatonin into our model. This was implemented with a triangle pulse function, pulsing for 12 hours a night, beginning at 9:00 pm and ending at 9:00 am each day.

Since we know the presence of melatonin can increase the period of *GnRH*, slowing down how often it pulses, we added a multiplier onto the period, that ranges from about 1 to 1.25.

From there, we included two multipliers, M_{FSH} and M_{LH} , applied to *FSH* and *LH* respectively. These are implemented with regards to melatonin as well, since melatonin increases *LH* and *FSH* sensitivity to *GnRH* pulses. These multipliers were estimated by data represented in [3] with a dose of 5 μg [3].

Since melatonin inhibits E_2 , we added a term, M_E , to account for this as well. This was determined through the data in [14], by plotting Estradiol against melatonin to observe how they relate. This behavior was then roughly fitted with a bell curve used to determine this multiplier.

In contrast, melatonin stimulates P_4 through the night, so that it is at its peak by morning, and we have included a term, M_{P_4} for this behavior.

4 Assumptions

Throughout our research process, the following statements were taken as assumptions. We first assume that the overall change of the hormones over the whole phase is negligible. Our model shows a direct impact of melatonin on *GnRH*, *LH*, *FSH*, P_4 and E_2 . We know that melatonin inhibits *GnRH* by inhibiting neuropeptide kisspeptin. For the purpose of our model, we applied melatonin's inhibitory effect directly to *GnRH*. To say that P_4 has a direct relationship with melatonin, we must assume that LH-induced StAR expression[2] is a separate pathway from melatonin's upregulation of StAR expression[5]. As we went through our research and literature review process we assumed that studies done on women undergoing in vitro fertilization treatment did not affect the information they provided. We also assumed that studies done on rats provided valid information about the dynamics of hormones, without taking into account precise values. Our last assumption is that endogenous and exogenous melatonin affect the system and the sex steroids in the same way.

5 Parameters and Equations

Parameters:

$k_{E_2} = 0.021$	Metabolic clearance rate of E_2
$k_{LH} = 0.07$	Metabolic clearance rate of LH
$k_{FSH} = 0.001$	Metabolic clearance rate of FSH
$k_{P_4} = 0.092$	Metabolic clearance rate of P_4

Time-dependent inputs:

$GnRH$	$GnRH$ PULSE function, built-in to Berkeley Madonna (adapted to Matlab)
mel	Triangular pulse function with period 1440 minutes, height 70 pg/mL, width 720 minutes

Variables:

$P = \begin{cases} 60 * P_{mel} & E_2 < 100 \\ P_{mel} * [(0.267 * E_2) - 6.75] & 100 \leq E_2 \leq 400 \\ 100 * P_{mel} & 400 < E_2 \end{cases}$	$GnRH$ period, dependent on E_2
$A = \begin{cases} 5 & E_2 < 100 \\ (-0.013 * E_2) + 8.2 & 100 \leq E_2 \leq 400 \\ 3 & 400 < E_2 \end{cases}$	$GnRH$ amplitude, dependent on E_2
$M = \begin{cases} 2 & E_2 < 128 \\ -0.002 * E_2 + 2.45 & 128 \leq E_2 \leq 800 \\ 0.5 & 800 < E_2 \end{cases}$	Multiplier for E_2 's inhibitory effect on LH

Time-dependent inputs:

$$mel_n = \begin{cases} 70 + \frac{7}{36} * (mod(t, 1440) - 180) & 180 \leq mod(t, 1440) \leq 540 \\ mel & \text{otherwise} \end{cases}$$

mel value dependent on time, amplifies melatonin value through the night

$$M_E = \begin{cases} 30 * e^{-\frac{(mel-40)^2}{800}} & mel \neq 0 \text{ and } 1260 \leq mod(t, 1440) \text{ or } mod(t, 1440) \leq 0 \\ 1 & \text{otherwise} \end{cases}$$

Multiplier for melatonin's stimulatory effect on E_2 at the start of the night

Melatonin-dependent variables:

$M_{FSH} = \begin{cases} 1.18 & mel \neq 0 \\ 1 & mel = 0 \end{cases}$	Multiplier for melatonin's stimulatory effect on FSH
$M_{LH} = \begin{cases} 1.193 & mel \neq 0 \\ 1 & mel = 0 \end{cases}$	Multiplier for melatonin's stimulatory effect on LH
$M_{P_4} = \begin{cases} \frac{0.01005}{140} * mel_n & mel \neq 0 \\ 1 & mel = 0 \end{cases}$	Multiplier for melatonin's stimulatory effect on P_4
$P_{mel} = \begin{cases} -\frac{(0.7+0.0001*mel)^2}{1+0.0001*mel^{2.1}} + 1.5 & mel \neq 0 \\ 1 & mel = 0 \end{cases}$	$GnRH$ period when melatonin is in the system, decreasing

Feedback functions:

$$\begin{aligned}
 FSH \rightarrow E_2 \text{ positive feedback} &= \frac{150 * (0.15 * FSH)^7}{1 + (0.15 * FSH)^7} + 250 \\
 E_2 \rightarrow FSH \text{ negative feedback} &= -\frac{9 * (0.0025 * E_2)^{15}}{1 + (0.0025 * E_2)^{15}} + 10 \\
 GnRH \rightarrow FSH \text{ positive feedback} &= \frac{5 * (0.2 * M_{FSH} * GnRH)^{15}}{1 + (0.2 * M_{FSH} * GnRH)^3} + 5
 \end{aligned}$$

Differential Equations:

$$\begin{aligned}
 \frac{dE_2}{dt} &= 0.0004 * M_E * (FSH \rightarrow E_2 \text{ positive feedback}) - (k_{E_2} * E_2) \\
 \frac{dFSH}{dt} &= 2 * 10^{-7} * [(E_2 \rightarrow FSH \text{ negative feedback}) + \\
 &\quad 0.1 * (GnRH \rightarrow FSH \text{ positive feedback})] - (k_{FSH} * FSH) \\
 \frac{dLH}{dt} &= (0.3 * M * M_{LH} * GnRH) - (k_{LH} * LH - 0.4) \\
 \frac{dP_4}{dt} &= (0.045 * LH) - (k_{P_4} * P_4) + (M_{P_4} * P_4)
 \end{aligned}$$

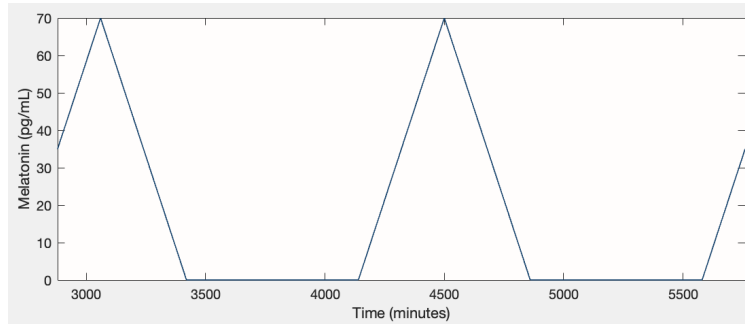


Figure 1: Melatonin pulse over the course of 2 days

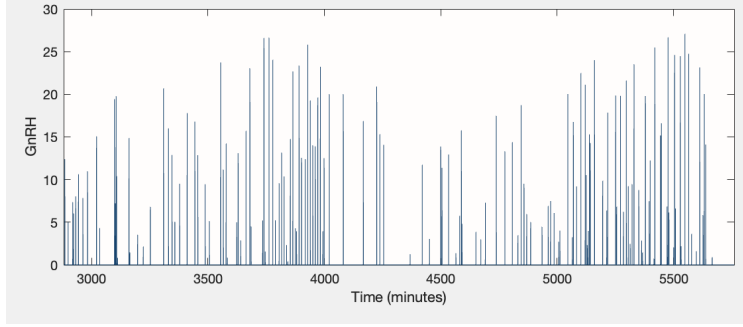


Figure 2: GnRH pulses over the course of 2 days

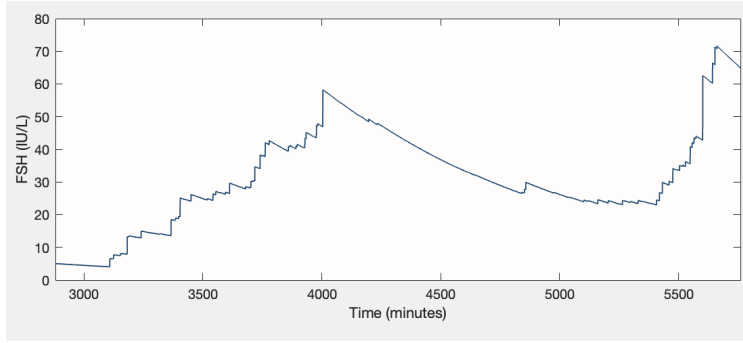


Figure 3: FSH over the course of 2 days

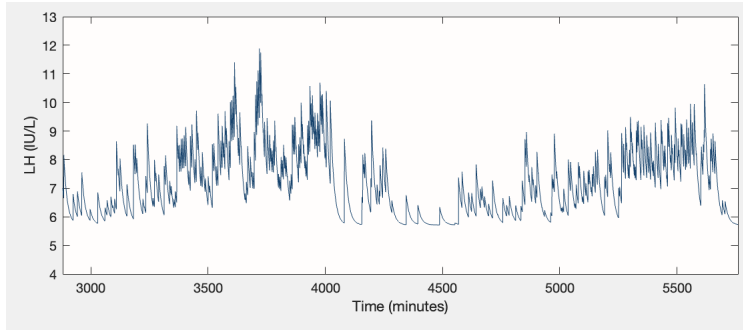


Figure 4: LH over the course of 2 days

6 Analysis

In our graphs, we can first note the direct impact that the level of E_2 has on $GnRH$, and how the periods and amplitudes are altered by changes in the values of E_2 (see Figure 2 and Figure 5). From there, we see how, under the influence of melatonin (Figure 1), LH and FSH follow a circadian rhythm. LH rises through the night, boosted by its increased sensitivity to $GnRH$'s pulses during the presence of melatonin (Figure 4). We can see how P_4 mirrors this behavior since it is stimulated by LH , and coupled with melatonin's stimulation of P_4 , we see how P_4 sharply rises

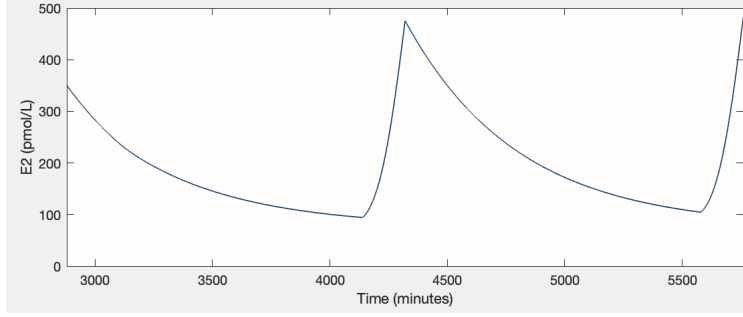


Figure 5: Estrogen over the course of 2 days

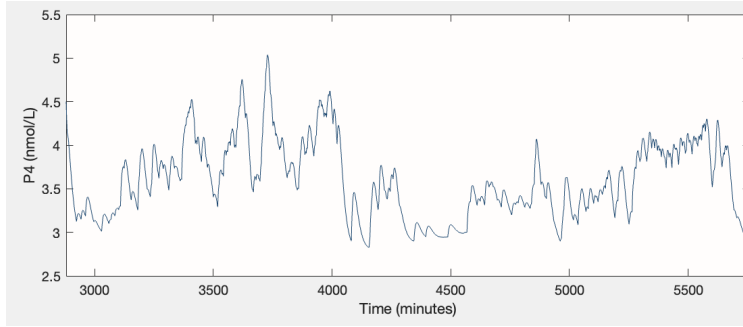


Figure 6: P4 over the course of 2 days

through the night and decreases through the day. On the other hand, FSH is much less sensitive to $GnRH$, so it is less affected by a boost in sensitivity, and instead is slightly lowered through the night due to the slowing of the frequency of $GnRH$ pulses (Figure 3).

We compared with data from [14], where we primarily focused on results from nine pre-menopausal women in their follicular phase. In this study, participants completed both Baseline and Constant Routine measurements in hope of observing a circadian rhythm for reproductive hormones. Plasma hormone levels were measured every 30 to 60 minutes. It was found that hormones more closely followed a rhythm during the follicular phase as opposed to the luteal phase. Graphs presented in [14] showed that LH and P_4 increase through the night, FSH remains relatively stable but decreases during the night, and E_2 decreases.

Our model appropriately represents this behavior, by observing melatonin's impact on each of the key reproductive hormones, we were able to implement these effects in our equations to obtain the desired fluctuations.

7 Limitations

Our set of equations is based on a model meant for the early follicular phase, while the graphs used to fit our model are from different stages of the follicular phase. This discrepancy has influenced the values of each hormone, and whether they are in the appropriate range. Another limitation that we faced during the project was the lack of studies done mapping out the circadian rhythm of

the menstrual cycle hormones during the follicular phase, while taking in to account the affects of melatonin. Lastly, it is difficult to obtain *GnRH* pulse data for humans as the pulses occur quickly and are not easy to capture. We instead used data from other species, such as rats, to make our assumptions.

8 Future Directions

Since our model shows the behavior of reproductive hormones on a minute-by-minute basis, it would be interesting to see how slight changes or abnormalities on a small scale impact larger systems. This could produce changes in the monthly menstrual cycle or even further assist in observing multi-system impacts, such as in the brain. In particular, Dr. Natalie Rasgon mentioned that women with irregular cycles may be more likely to develop mood disorders, such as bipolar disorder. Observed irregularity in menstrual cycles and hormone fluctuation may be an indicator of a future mood disorder diagnosis. Analyzing this behavior on the smaller scale could help us see how daily behaviors impact long-term health complications.

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References

- [1] Fiona C. Baker and Helen S. Driver. “Circadian rhythms, sleep, and the menstrual cycle”. In: *Sleep Medicine* 8.6 (2007). Circadian Rhythms in Sleep Medicine, pp. 613–622. ISSN: 1389-9457. DOI: <https://doi.org/10.1016/j.sleep.2006.09.011>. URL: <https://www.sciencedirect.com/science/article/pii/S1389945706006216>.
- [2] J. Bódis et al. “Influence of Melatonin on Basal and Gonadotropin-Stimulated Progesterone and Estradiol Secretion of Cultured Human Granulosa Cells and in the Superfused Granulosa Cell System”. In: *Gynecologic and Obstetric Investigation* 52.3 (Oct. 2001), pp. 198–202. ISSN: 0378-7346. DOI: 10.1159/000052973. eprint: <https://karger.com/goi/article-pdf/52/3/198/2865541/000052973.pdf>. URL: <https://doi.org/10.1159/000052973>.
- [3] A Cagnacci et al. “Melatonin enhances the luteinizing hormone and follicle-stimulating hormone responses to gonadotropin-releasing hormone in the follicular, but not in the luteal, menstrual phase”. In: *The Journal of Clinical Endocrinology Metabolism* 80.4 (Apr. 1995), pp. 1095–1099. ISSN: 0021-972X. DOI: 10.1210/jcem.80.4.7714075. eprint: <https://>

- academic.oup.com/jcem/article-pdf/80/4/1095/10505496/jcem1095.pdf. URL: <https://doi.org/10.1210/jcem.80.4.7714075>.
- [4] Christian O. Casteel and Gurdeep Singh. *Physiology, Gonadotropin-Releasing Hormone*. StatPearls Publishing, Treasure Island (FL), 2023. URL: <http://europepmc.org/books/NBK558992>.
 - [5] Lanlan Fang et al. “Melatonin induces progesterone production in human granulosa-lutein cells through upregulation of StAR expression”. In: *Aging* 11 (Oct. 2019). DOI: 10.18632/aging.102367.
 - [6] N Friis Wang et al. “Response to ovulation trigger is correlated to late follicular phase progesterone levels: A hypothesis explaining reduced reproductive outcomes caused by increased late follicular progesterone rise”. In: *Human Reproduction* 34.5 (Mar. 2019), pp. 942–948. ISSN: 0268-1161. DOI: 10.1093/humrep/dez023. eprint: <https://academic.oup.com/humrep/article-pdf/34/5/942/28570621/dez023.pdf>. URL: <https://doi.org/10.1093/humrep/dez023>.
 - [7] Janet E. Hall. “Chapter 7 - Neuroendocrine Control of the Menstrual Cycle”. In: *Yen and Jaffe’s Reproductive Endocrinology (Eighth Edition)*. Ed. by Jerome F. Strauss and Robert L. Barbieri. Eighth Edition. Philadelphia: Elsevier, 2019, 149–166.e5. ISBN: 978-0-323-47912-7. DOI: <https://doi.org/10.1016/B978-0-323-47912-7.00007-X>. URL: <https://www.sciencedirect.com/science/article/pii/B978032347912700007X>.
 - [8] Leona Harris Clark, Paul M. Schlosser, and James F. Selgrade. “Multiple Stable Periodic Solutions in a Model for Hormonal Control of the Menstrual Cycle”. In: *Bulletin of Mathematical Biology* 65.1 (2003), pp. 157–173. ISSN: 0092-8240. DOI: <https://doi.org/10.1006/bulm.2002.0326>. URL: <https://www.sciencedirect.com/science/article/pii/S0092824002903265>.
 - [9] Pedro Marques et al. *Physiology of GnRH and Gonadotropin Secretion*. MDText.com, Inc., South Dartmouth (MA), 2000. URL: <http://europepmc.org/books/NBK279070>.
 - [10] Alexander Medvedev, Anton V. Proskurnikov, and Zhanybai T. Zhusubaliyev. “Mathematical modeling of endocrine regulation subject to circadian rhythm”. In: *Annual Reviews in Control* 46 (2018), pp. 148–164. ISSN: 1367-5788. DOI: <https://doi.org/10.1016/j.arcontrol.2018.08.002>. URL: <https://www.sciencedirect.com/science/article/pii/S1367578818300385>.
 - [11] Charles Meliska, Luis Martínez, and Barbara Parry. “Sleep—endocrine relationships in depressed women across the reproductive cycle”. In: *Sleep and Mental Illness* (Jan. 2010), pp. 101–118. DOI: 10.1017/CB09781139042734.012.
 - [12] Anna Möller and Torbjorn Backstrom. “Diurnal Variations of Endogenous Steroids in the Follicular Phase of the Menstrual Cycle”. In: *Neurochemistry Neuropharmacology* 2 (Jan. 2016). DOI: 10.4172/2469-9780.1000109.

- [13] Shadab A Rahman et al. “Endogenous Circadian Regulation of Female Reproductive Hormones”. In: *The Journal of Clinical Endocrinology Metabolism* 104.12 (Aug. 2019), pp. 6049–6059. ISSN: 0021-972X. DOI: 10.1210/jc.2019-00803. eprint: <https://academic.oup.com/jcem/article-pdf/104/12/6049/30312367/jc.2019-00803.pdf>. URL: <https://doi.org/10.1210/jc.2019-00803>.
- [14] Shadab A Rahman et al. “SUPPLEMENTAL INFORMATION TO RAHMAN ET AL.docx”. In: (Sept. 2019). DOI: 10.6084/m9.figshare.8214779.v1. URL: https://figshare.com/articles/figure/SUPPLEMENTAL_INFORMATION_TO_RAHMAN_ET_AL_docx/8214779.
- [15] Natalie L. Rasgon et al. “Emergent Oscillations in Mathematical Model of the Human Menstrual Cycle”. In: *CNS Spectrums* 8.11 (2003), pp. 805–814. DOI: 10.1017/S1092852900019246.
- [16] Ari Shechter and Diane B. Boivin. “Sleep, Hormones, and Circadian Rhythms throughout the Menstrual Cycle in Healthy Women and Women with Premenstrual Dysphoric Disorder”. In: *International Journal of Endocrinology* 2010.1 (2010), p. 259345. DOI: <https://doi.org/10.1155/2010/259345>. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1155/2010/259345>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1155/2010/259345>.
- [17] Turek FW Vitaterna MH Takahashi JS. “Overview of circadian rhythms.” In: *The Journal of Clinical Endocrinology Metabolism* 25.2 (2001), pp. 85–93. URL: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6707128/>.